Project Report

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

ServiceNow Flow Designer

Team ID	SWTID-2026-1951
Team Member 1	Nancy.D
Team Member 2	Mythely.J
Team Member 3	Srimathi.K
Team Member 4	Sahasri.C

Contents

1. Introduction
2. Ideation Phase
3. Requirement Analysis
4. Project Design
5. Project Planning & Scheduling
6. Functional and Performance Testing
7. Results
8. Advantages & Disadvantages
9. Conclusion
10. Future Scope
11. Appendix

1. INTRODUCTION

1.1 Project Overview

In many organisations an employee who needs a standard laptop raises a request that is routed by email or by hand through managers, procurement and IT. Each hand-off is a chance for delay, lost requests and inconsistent data. This project uses ServiceNow Flow Designer to automate the complete lifecycle of a standard laptop order: the employee submits a Service Catalog request, a flow gathers the details, obtains manager approval, creates the procurement task, keeps the requester informed and closes the request once the laptop is delivered and recorded as an asset.

1.2 Purpose

- Reduce the manual effort IT and procurement spend on routine laptop requests.
- Shorten approval and fulfilment time with rule-based routing and automatic reminders.
- Give requesters real-time status updates and give managers a single approval point.
- Create a consistent, auditable record of every order from request to delivery.
- Demonstrate low-code automation on the ServiceNow platform using Flow Designer.

2. IDEATION PHASE

2.1 Problem Statement

How might we automate the ordering of standard laptops so that employee requests are approved, fulfilled and tracked automatically in ServiceNow, reducing manual effort, approval delays and procurement errors for the IT team?

Problem Statement	I am	I'm trying to	But	Because	Which makes me feel
PS-1	an employee needing a laptop	get a standard laptop quickly	I do not know who is handling my request or how long it will take	requests move through emails and manual hand-offs	frustrated and unproductive
PS-2	an IT procurement analyst	process laptop orders accurately	I re-enter the same details and chase approvals	there is no automated workflow	overloaded and error-prone

2.2 Empathy Map Canvas

SAYS

- "Where is my laptop request?"
- "Why does approval take so long?"

THINKS

- Will I get my laptop on time?
- Is my request stuck somewhere?

DOES

- Emails IT and manager for updates
- Re-submits missing details

FEELS

- Frustrated by delays
- Uncertain about status

Figure 2.2: Empathy map for the employee requesting a laptop (and the IT procurement team)

2.3 Brainstorming

The team (Nancy.D, Mythely.J, Srimathi.K and Sahasri.C) gathered, agreed the problem statement above and listed ideas individually before grouping them. The ideas were then prioritised by importance and feasibility.

Theme	Ideas	Priority
Request Intake	Standard laptop catalog item; auto-fill user, department and location; trigger flow on submission	High
Approval Workflow	Manager approval action; auto-approve under a cost threshold; escalate approvals pending for 2 days	High
Fulfilment & Tracking	Procurement task for vendor order; update asset record on delivery; dashboard of open requests	Medium to High
Governance & Reliability	Notifications to requester; error handling and retry; audit log of approvals	Medium

The full brainstorm and prioritisation grid is provided in the separate Brainstorm & Idea Prioritization document.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Need a laptop	Submit request	Approval	Procurement	Delivery	Closure
Actions	Identifies need; opens Service Catalog	Selects standard laptop item and submits	Manager reviews request	IT creates vendor order	Receives and sets up laptop	Confirms receipt
Touchpoint	Employee portal	Catalog form	Approval notification	Procurement task	Asset record	Request state
Pain point (before)	Unclear process	Re-typing details	Chasing approvers	Manual follow-up	No visibility	Unclosed tickets
Automation (after)	Single catalog entry	Auto-filled form; flow triggered	Auto-routed approval with reminders	Task auto-created	Status e-mails sent	Auto-close and asset update

Figure 3.1: Customer journey map, from need to closure

3.2 Solution Requirement

Functional requirements

ID	Requirement	Description
FR-1	Catalog request	Employee submits a standard laptop request through a Service Catalog item with user details auto-filled.
FR-2	Flow trigger	Flow Designer flow starts automatically when the request item is created.
FR-3	Approval	Manager receives an approval request; approve or reject updates the request state.
FR-4	Procurement task	On approval, a procurement task is created and assigned to the procurement group.
FR-5	Notifications	Requester is notified at submission, approval, rejection and delivery.
FR-6	Closure and asset	On delivery the request is closed and the laptop is recorded as an asset.

Non-functional requirements

ID	Requirement	Description
NFR-1	Usability	Request form is short and understandable without training.
NFR-2	Security	Only managers can approve; access controlled by ServiceNow roles.
NFR-3	Reliability	Flow handles failures with error handling and retries.
NFR-4	Auditability	Every approval and state change is recorded in the activity log.
NFR-5	Maintainability	Logic is built in low-code Flow Designer so it can be changed without scripting.

3.3 Data Flow Diagram

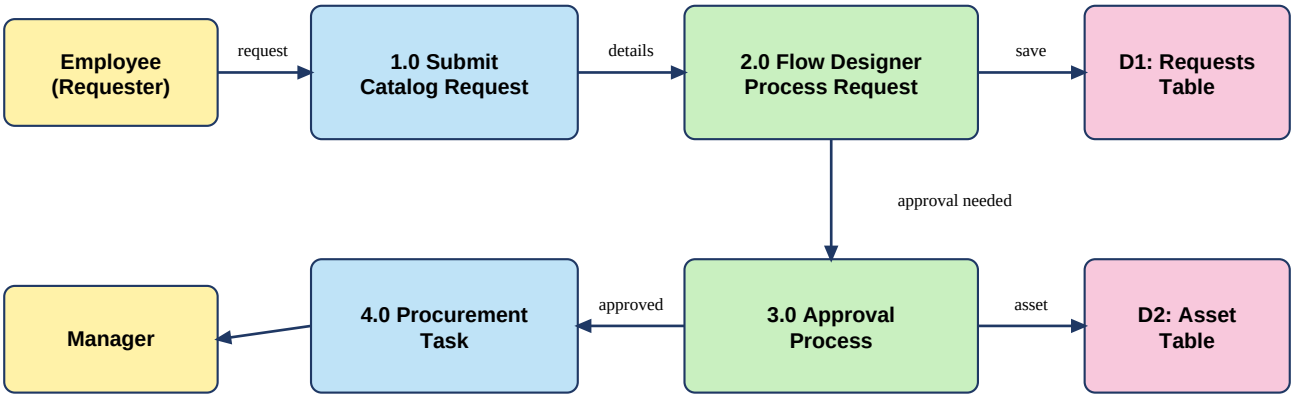

Figure 3.3: Level-1 data flow diagram of the laptop ordering process

3.4 Technology Stack

Layer	Technology
Platform	ServiceNow (Personal Developer Instance)
Automation	Flow Designer: flows, actions, approvals, conditions
Request entry	Service Catalog item and variables
Data	ServiceNow tables: Requested Item, Catalog Task, Asset
Notifications	ServiceNow notifications and email actions
Access control	Roles and groups: requester, manager, procurement

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Proposed fit
Requests tracked by email and hand-offs	A single catalog item starts one automated flow
Approvals delayed or forgotten	Approval action with reminders and escalation
Same details re-entered by IT	Variables auto-filled and passed through the flow
No status visibility for requester	Notifications at every state change
Weak audit trail	Activity log and asset record for each order

4.2 Proposed Solution

Parameter	Description
Problem being solved	Slow, manual and error-prone ordering of standard laptops
Idea / solution	Flow Designer flow triggered by a catalog request that handles approval, procurement task, notifications and closure
Novelty	Fully low-code process with threshold-based auto-approval and escalation
Social impact	Employees receive equipment faster and IT staff are freed for higher-value work
Business model	Internal efficiency gain: fewer manual hours per order and a lower error rate
Scalability	The same pattern can be reused for monitors, phones and software requests

4.3 Solution Architecture

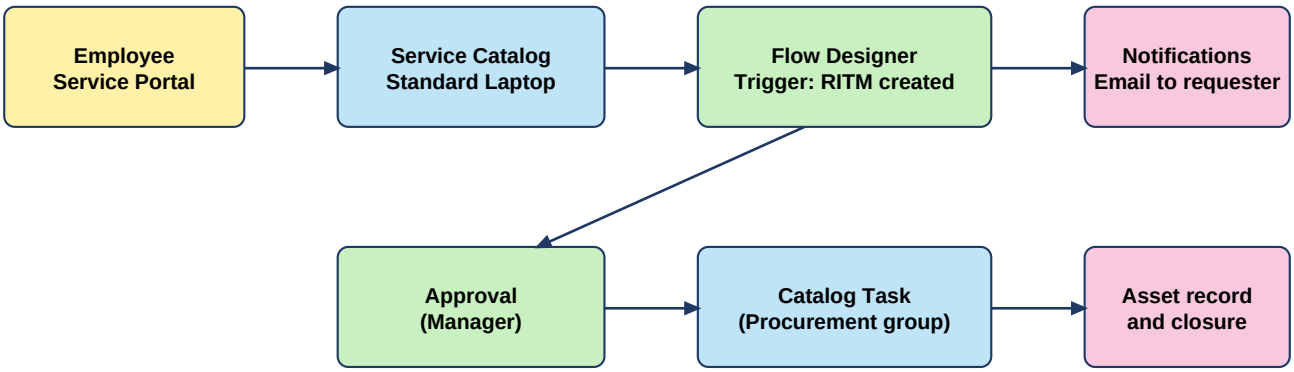

Figure 4.3: Solution architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Task	Owner	Outcome
Sprint 1	Problem definition, empathy map, brainstorming	Whole team	Problem statement and prioritised ideas
Sprint 1	Requirement analysis, journey map, data flow diagram	Nancy.D, Mythely.J	Requirements and diagrams documented
Sprint 2	Create catalog item and variables	Srimathi.K	Standard laptop item available in catalog
Sprint 2	Build flow: trigger, approval, procurement task	Nancy.D, Sahasri.C	Working end-to-end flow
Sprint 3	Notifications, error handling, asset update	Mythely.J, Srimathi.K	Complete automation
Sprint 3	Testing, screenshots and report	Whole team	Tested solution and final report

Owners and sprint split are a proposed plan; adjust to match the team's actual allocation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The following scenarios are used to verify the flow. The Result column is left to be completed from the team's actual test run.

ID	Scenario	Expected result	Result
T1	Submit standard laptop request	Request item created and flow starts	To be recorded
T2	Manager approves	Procurement task created and requester notified	To be recorded
T3	Manager rejects	Request closed as rejected and requester notified	To be recorded
T4	Approval pending beyond limit	Reminder or escalation sent	To be recorded
T5	Order below cost threshold	Auto-approved without manual step	To be recorded
T6	Procurement task completed	Asset record updated and request closed	To be recorded
T7	Failure in a flow action	Error is logged and action retried	To be recorded

Performance metrics tracked: time from submission to approval, time from approval to task creation, flow execution time (from Flow Designer execution details) and number of failed executions.

7. RESULTS

7.1 Output Screenshots

Service Catalog item: Standard Laptop

Flow Designer flow canvas

Approval request view

Procurement catalog task

Flow execution details

Notification email / closed request

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Removes repetitive manual work for IT and procurement	Limited to standard laptop configurations
Faster, consistent approvals with reminders	Needs correct approver and group data to route properly
Requester always informed of status	Complex exceptions still need manual handling
Full audit trail of every order	Dependent on ServiceNow licensing and platform availability
Low-code, so easy to change and extend	No live integration with an external vendor ordering system yet

9. CONCLUSION

The project shows that ServiceNow Flow Designer can turn a slow, email-driven laptop ordering process into a single automated workflow covering request, approval, procurement, notification and closure. The approach reduces manual effort and delay, improves visibility for employees and gives IT a reliable audit trail, using only low-code configuration.

10. FUTURE SCOPE

- Extend the pattern to other standard equipment such as monitors, phones and accessories.
- Integrate with a vendor or ERP system so purchase orders are placed automatically.
- Add budget checks and role-based thresholds for auto-approval.
- Add a self-service dashboard and reports on request volume and cycle time.
- Offer a virtual agent in the portal so employees can order and track by chat.

11. APPENDIX

Item	Details
Source Code (if any)	ServiceNow update set / flow export: to be added
Dataset Link	Not applicable (data held in ServiceNow tables)
GitHub & Project Demo Link	To be added